Smaller Diophantine Equations for ZFC Consistency

Four-root separation, normalized radix positions, and direct certificates with recovered domains

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AI-assisted research by ChatGPT; proofs and exact artifacts supplied for review

23 October 2026
Comparison snapshot: `7e533c33819ee1f703b69f2778a767cdc069940d`

Abstract

We give seven explicit ordinary Diophantine polynomials, each with a two-way existence reduction to a fixed syntactic refutation predicate for ZFC. The principal unrestricted-integer proposal uses fourteen unknowns, has exact degree 540, and admits a 160-operation arithmetic circuit. A 147-operation companion has degree 568. Three positive-domain equations retain nine unknowns while improving their immediate predecessors in degree and circuit length; the shortest has 420 operations. A separate direct proof-calculus construction gives a quartic in 495 nonnegative unknowns. Recovering most coordinate signs from its own equations reduces the unrestricted-integer lift to 977 unknowns, with 24,632 explicitly listed monomials and coefficient height 83,869,250.

The new arguments are ordinary mathematical proofs composed with pinned ProveIt calibration theorems. Exact finite checks reconstruct coefficients, replay transformations, establish degree certificates, and reject corrupted data. We supply neither a new end-to-end proof-assistant theorem nor an object-language ZFC proof file. No global minimality or historical priority is claimed. The contribution is a reproducible improvement of several distinct measures within the inspected repository, together with structural ideas that can be developed further.

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1 The result and what "simplest" measures

An ordinary Diophantine equation is a polynomial equation with fixed integer coefficients. Its unknowns range over a specified integer domain. Every exponent in such a polynomial is a fixed nonnegative integer; a variable exponent, factorial, or binomial symbol cannot remain as a primitive in the final equation. All seven delivered artifacts meet this requirement. Registers in the compact formulas denote polynomial abbreviations. They are not extra existential unknowns.

There are several conflicting measures of simplicity: the number of unknowns, total degree, arithmetic-circuit length, coefficient magnitude, and the size of the correctness argument. A circuit can be very short while referring to large fixed coefficients. An expanded quartic can have modest coefficients while needing many variables. The same distinction underlies the study of universal variable–degree pairs [4], although the present problem concerns a single fixed consistency instance rather than a universal family.

Equation	Domain	Unknowns	Exact degree	Further measure
Boundary, simple	$\mathbb{Z}_{>0}$	9	1,472,168	420 operations
Boundary, complete	$\mathbb{Z}_{>0}$	9	997,810	1,057 operations
Compact, simple	$\mathbb{Z}_{>0}$	9	1,626,248	428 operations
P_{540}	$\mathbb{Z}$	14	540	160 operations
P_{568}	$\mathbb{Z}$	14	568	147 operations
P_{495}	$\mathbb{Z}_{\geq 0}$	495	4	15,377 monomials
P_{977}	$\mathbb{Z}$	977	4	24,632 monomials

Table 1: The seven delivered polynomials. The domain is part of each claim.

For unrestricted integer variables with moderate degree and a short formula, our main proposal is P_{540}. Its lower-operation companion P_{568} provides a useful alternative. If positive integers are the intended domain, the 420-operation equation is the shortest new evaluator in this packet. The compact 428-operation form keeps a smaller fixed coefficient tuple. If a direct proof-calculus source and an entirely expanded polynomial with moderate coefficients are the priority, use P_{977}.

The comparison is deliberately specific. At the inspected commit [1], incoming fourteen-integer forms included degree 652 with 230 operations, degree 696 with 150 operations, and degree 792 with 137 operations. Thus $(568, 147)$ improves $(696, 150)$ in both measurements, and $(540, 160)$ offers a still lower degree. The degree-792 form retains a smaller circuit. Other incoming constructions use fewer than fourteen unrestricted integers, at much higher degree. The new fourteen-variable equations are not variable-count records.

The direct comparison is particularly large: the synchronized parent has a quartic lift in 2,088 unrestricted integers. Our final lift has 977, a reduction of 1,111 variables, or approximately 53.2%. Its coefficient height is also slightly smaller, and the quartic degree is unchanged. These numbers compare actual collected equations under the same domain, not a natural-domain equation against an integer-domain one.

1.1 The precise logical claim

Fix the repository's sealed recursive presentation of the ZFC axioms and its finite first-order derivation predicate. Write

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Refutation}_{\mathrm{ZFC}}(p).$$

This notation concerns syntactic consistency. It is not an assumption that every theorem of ZFC is true in an intended universe.

Theorem 1.1 (Ordinary consistency equivalence)**.** *For each delivered polynomial P and its domain D^n in Table 1, the new reductions and the named inherited calibration theorems give an ordinary proof formalizable in ZFC of*

$$\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \exists p \in \mathbb{N} \; \mathrm{Refutation}_{\mathrm{ZFC}}(p).$$

Consequently,

$$\boxed{\mathrm{ZFC} \vdash (\nexists \mathbf{x} \in D^n : \; P(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).}$$

The proof is compositional: Sections 2, 3, 4, and 5 prove the new existence equivalences; Section 6 identifies the imported endpoints. None of these implications assumes consistency. In particular, the theorem does not assert the absence of solutions. A proof of that further assertion inside ZFC would also prove its consistency; the present biconditional does not supply such a proof.

1.2 Three ideas responsible for the improvements

Separate the guard before combining radicals. The fourteen-integer equation keeps one square-root witness explicit. Only four Pell radicals remain in the norm. Their magnitudes already admit a strict superincreasing order under a weak scale hypothesis. That observation removes a large auxiliary amplifier. Rational extraction can therefore precede the CRT argument that restores the stronger cubic scale. Centering the norm around polynomial root approximations gives the degree-540 form.

Use the units digit of the coefficient layout. The positive nine-variable interface does not require an empty radix position zero. A uniform origin translation preserves every source coefficient identity, lowers the convolution height, and permits a new exact population-count calibration. Shared powers then shorten the literal evaluator. This is a reduction in a valid encoding, not merely a change in the formatting of a polynomial.

Recover domains from the arithmetic certificate. The direct quartic already forces most of its coordinates to be nonnegative. The common Pell frame supplies additional signs using only ordered-ring inequalities, before invoking its stronger power semantics. Only 170 coordinates need explicit nonnegativity representations. Applying signed eliminations afterward yields the 977-integer equation. This order prevents the usual independent representation of every natural coordinate from dominating the variable count.

2 Four Pell roots: fourteen integers at degrees 540 and 568

This section gives two ordinary polynomials whose fourteen coordinates all range over the unrestricted integers. The first has exact degree 540 and 160 live binary arithmetic operations; the second has degree 568 and 147 operations. The fixed coefficient tuple has 615 883 raw minimal unsigned bytes, with the original six integer payloads retained. Operations count shared binary addition, subtraction, and multiplication, including multiplication by a fixed constant. A register is a polynomial abbreviation, not an additional quantified coordinate.

2.1 Literal definition

The variables, in circuit order, are

$$j, C, z, h, r, v, w, T, c, n, t, p_2, p_3, p_1 \in \mathbb{Z}.$$

Use the six fixed integers of the retained source-34/source-71 tuple:

$$\beta = 2^{1064921} - 1, \quad K_0, M_C, M_F, \quad A_{\mathrm{c}} = 2^{1113}, \quad B_{\mathrm{c}} = 2^{666104}.$$

Here M_F is already shifted by β. The exact integer bytes, finite machine-and-tile recipe, original licenses, and ordinary calibration texts are included in the component's `upstream/` directory. This tuple is larger than some other native tuples in ProveIt; the present determinant scales are proved for this whole tuple and are not mixed with another one.

Define

$$\begin{aligned} J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4, \\ X &= qW, & Y &= qv, & E &= XY, \\ a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H. \end{aligned}$$

Then define the transport and rank registers

$$\begin{aligned} F &= (K_0 + W)C - z(q - 1), \\ R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J. \end{aligned}$$

Continue with

$$\begin{aligned} k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta, \\ i &= 2r + 1, & I &= ic^2, & D_f &= 1 + \Delta I^2, \\ Q &= \Delta^2 I^2, & \delta &= Q - 1, & V &= D_f(cT - R) - c, \\ C_y &= QV^2 - 1, & \ell &= XY^2, & L &= \ell k, \\ D_\tau &= 1 + \ell(\ell + 1)k^2, & D_m &= 1 + \Delta c^2, & & \\ b_0 &= X + ac, & F_0 &= (a + 2)I. & & \end{aligned}$$

Put

$$\begin{aligned} \mathrm{Pos}(u_1, u_2, u_3) &= 1 + u_1^2 + u_2^2 + u_3(u_3 + 1), \\ G &= (A_{\mathrm{c}}F)(B_{\mathrm{c}}C) - (q - A_{\mathrm{c}}F - B_{\mathrm{c}}C)^2, \\ B &= v^2 G\eta\zeta - (t^2 - v^2 G - \eta\zeta + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3), \qquad S = B^2. \end{aligned}$$

Let

$$\mathcal{F}_{16}(x; A_1, A_2, A_3, A_4) = \prod_{\varepsilon \in \{-1,1\}^4} \left(x - \sum_{j=1}^{4} \varepsilon_j \sqrt{A_j} \right).$$

Sign invariance makes this an integer polynomial in its displayed arguments. Its exact 495 nonzero coefficients are verified against the independent product of the sixteen factors.

For the centered equation define

$$\begin{aligned} M &= 4q^2, & U &= MH, \\ Z &= U(n + 2V + 2L + k + 2F_0) + 2Mb_0 + 2H(v - q^2), \\ N &= \delta^8 \mathcal{F}_{16}\left(Z; 4U^2 D_f, 4U^2 D_\tau, 4M^2 D_m, \frac{4U^2 C_y}{\delta} \right). \end{aligned}$$

$$\boxed{P_{540} = N + S + NS = 0.}$$

The denominator is cleared symbolically; the literal evaluator contains only integer polynomial operations. For the shorter companion use instead

$$M = 2q^2, \qquad U = MH,$$
$$Z = U(n+V) + Mb_0 + H(v-q^2),$$
$$N = \delta^8 \mathcal{F}_{16}\left(Z; U^2D_f, U^2D_\tau, M^2D_m, \frac{U^2C_y}{\delta}\right).$$

$$\boxed{P_{568} = N + S + NS = 0.}$$

All other registers remain unchanged.

2.2 The two guard facts

For integers x, y, t define

$$\mathrm{Pair}(x,y,t) = xy - (t^2 - x - y + 1)^2.$$

Then

$$(\exists t \in \mathbb{Z})\ \mathrm{Pair}(x,y,t) > 0 \quad \Longleftrightarrow \quad x > 0 \text{ and } y > 0.$$

The forward direction follows because positive Pair forces $xy > 0$, whereas for $x = -u, y = -v$ with $u, v > 0$ its subtracted square is at least $(u+v+1)^2 > uv$. For the reverse direction set $K = x + y - 1$ and choose a nearest integer square t^2. Between consecutive squares $m^2, (m+1)^2$ the error is at most m, so $(t^2 - K)^2 < K \le xy$; square endpoints, including $K = 1$, have zero error.

The polynomial Pos represents precisely the positive integers. Positivity is immediate; surjectivity follows from the three-square theorem applied to $4n - 3$, with one odd and two even square coordinates. Consequently, $B = 0$ forces $v \neq 0$, $G > 0$, and $\eta\zeta > 0$. Conversely these three conditions admit private witnesses making $B = 0$.

Since $S = B^2 \ge 0$, the equation

$$P = (1+S)(1+N) - 1 = 0$$

forces $S = N = 0$: $1 + S$ is a positive integer divisor of 1. No sign hypothesis about the norm N is used.

2.3 Rational extraction before the cubic scale is known

The front end gives $q \equiv 2 \pmod 6$ and $|q| \ge 16$. Also $W = 4(3w+1) \neq 0$, hence $x = |X| \ge 64$. When $v = 0$ we have $a = 0$, $H = 3$, $\Delta = 3$; when $v \neq 0$, $m = |Y| \ge 16$ and $|a| \ge 1008$. Thus $H \neq 0$ and $\Delta \ge 3$ on the entire integer domain. The three integer radicands D_m, D_f, D_τ are positive, the last by $\ell(\ell+1) \ge 0$.

If $c = 0$, then $Q = 0$, $\delta = -1$, and $C_y/\delta = 1$. If $c \neq 0$, then $Q \ge 9$, $D_f > |c| + 1$, and the integer $V = D_f(cT - R) - c$ has $|V| \ge 1$. Therefore

$$\frac{C_y}{\delta} = V^2 + \frac{V^2 - 1}{Q - 1} \ge 1.$$

In particular δ never vanishes, including all exceptional branches.

At a polynomial zero, the guard supplies $v \neq 0$, $G > 0$, and $\eta\zeta > 0$. Write $K = |k|$, $C_0 = |c|$. Then $K \ge 2$ and $0 < \eta/k < 1$, so

$$K(m-1) < C_0 < K(m+1), \qquad C_0 > 30.$$

Let d, f, τ, y be the positive square roots of $D_m, D_f, D_\tau, C_y/\delta$, respectively. They satisfy

$$\frac{d}{|H|} < \frac{C_0}{2}, \qquad \tau > 2C_0, \qquad f > 2\tau.$$

For the first inequality use $H^2 - 4\Delta = 12a^2 + 8a - 3$. For the second use $K(xm^2 - 1) < \tau < K(xm^2 + 1)$. For the third,

$$d > ((x-1)m - 3)K(m-1) > \frac{3}{4}\tau, \qquad f > \frac{C_0}{2}d > 15d.$$

The middle comparison follows by expanding $m^2(x-4) - 4m(x+2) + 9 > 0$ for $x \geq 64, m \geq 16$. Also $f > 3C_0^2$ and $Q > f^2$.

If $cT - R = 0$, then $C_0 < y < C_0 + 1$. Otherwise $y > |V| \geq f^2 - C_0 > 2f$. Hence either

$$f, \tau, y, d/|H| \quad \text{or} \quad y, f, \tau, d/|H|$$

is a strictly superincreasing descending list: every entry exceeds the sum of all later entries. For example, $y + d/|H| < 1.5C_0 + 1 < 2C_0 < \tau$ in the first case. The deleted source amplifier $4H + 1$ is unnecessary because the guard root is now a separate explicit coordinate, leaving these four roots to separate.

A rational signed sum of positive roots with rational squares and strictly superincreasing magnitudes makes every root rational. Indeed, a nontrivial Galois automorphism changing at least one radical sign would yield a nonempty vanishing signed subsum after subtraction, contradicting the largest-term inequality. This argument allows dependent square classes. At $N = 0$ one conjugate factor vanishes; division by $2U$ in the centered case, or U in the companion, gives exactly such a rational sum. Thus d, f, τ are integers.

The auxiliary rational y is also integral. For any integers $Q > 1, V$,

$$QV^2 - (Q-1)y^2 = 1, \quad y \in \mathbb{Q} \quad \Longrightarrow \quad y \in \mathbb{Z}.$$

If $y = a/b$ is reduced, then $b^2 \mid Q - 1$. The transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves the equation and denominator, keeps V' integral, and for $V > 1$ gives $-V < V' < V$. Here $y > V$ and $(Q-1)y < QV$ give the two strict inequalities. The equation excludes $V' = 0$. Descent reaches $|V| = 1$, where $|y| = 1$, proving the claim.

2.4 The congruences and the consistency equivalence

Absorb the conjugate signs into integer roots. In the centered case their relation is

$$Z = 2Md + 2U(f + \tau + y).$$

Since $H \equiv 3 \pmod q$, $q \equiv 2 \pmod 3$, and H is odd, $\gcd(H, 4q^2) = 1$. Reduction modulo $M = 4q^2$ yields $2q^2 \mid v - q^2$, and reduction modulo H yields $H \mid d - b_0$. Therefore

$$v = q^2(2v_{\text{old}} + 1), \quad Y = (2v_{\text{old}} + 1)q^3, \quad \gamma = (d - b_0)/H \in \mathbb{Z}.$$

The companion with $M = 2q^2$ gives the same conclusions without the factors 2. No cubic-scale promise was used in the preceding extraction.

These roots, the restored scale, and the two strict guards give the retained parent's normalized signed kernel. Its original strong variables can be restored explicitly:

$$I_{\text{old}} = \Delta i c^2, \qquad r_{\text{old}} = (\Delta i - 1)/2,$$
$$o_{\text{old}} = f(cT - R), \qquad j_{\text{old}} = D_f T - 1 - R\Delta i^2 c^3.$$

The quotient is integral because Δi is odd. Substitution gives $I_{\mathrm{old}}^2 = \Delta(f^2-1) = Q$ and $fo_{\mathrm{old}} - c = cj_{\mathrm{old}} - R = V$. Private old guard witnesses are regenerated from $G > 0, \eta\zeta > 0$.

Conversely, take a normalized source witness, set $v = q^2(2v_{\mathrm{old}}+1)$, and choose the private guard witnesses by Pair and Pos. For the centered equation choose

$$n = 2(\gamma + f + \tau + y - V - L - F_0) - k - \frac{v-q^2}{2q^2}.$$

For the companion choose

$$n = \gamma + f + \tau + y - V - \frac{v-q^2}{2q^2}.$$

Both are integers and make the selected conjugate factor vanish. Every quantity is chosen before n, so the construction is not circular.

We have proved an exact two-way existence reduction to the normalized kernel. Composing with the parent's explicitly identified ordinary native compiler and K215/refutation calibration yields, for $d = 540,568$,

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{14}\ P_d(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}.$$

This imports ordinary mathematical theorems and their stated canonical completeness. It does not promote finite arithmetic checks into an unbounded semantic proof, or supply the separately open encoded-ZFC proof artifact.

2.5 Exact degrees, including cancellation

Treat the fixed coefficients as degree zero. For leading parts set

$$\begin{aligned}
&q_0 = 6\beta j, && W_0 = 12w, && X_0 = q_0W_0, && Y_0 = q_0v,\\
&a_0 = X_0Y_0, && i_0 = 2r, && R_0 = q_0^4z - q_0^3W_0C,\\
&k_0 = R_0 + ha_0, && \eta_0 = -Y_0k_0, && Q_0 = a_0^4i_0^2c^4,\\
&V_0 = -a_0^2i_0^2c^4R_0, && L_0 = X_0Y_0^2k_0, && F_{00} = a_0i_0c^2.
\end{aligned}$$

The degrees of $\delta, U, V, L, F_0, \eta$ are $22, 6, 19, 11, 7, 7$. The guard has top part $-\eta_0^4$, so S has degree 56 and leading part η_0^8.

For the centered equation use approximate roots

$$r_y = 2UV, \quad r_\tau = U(2L+k), \quad r_f = 2UF_0, \quad r_d = 2Mb_0, \quad s = Un + 2H(v-q^2).$$

The center is their sum plus s. The four exact square defects are

$$\begin{aligned}
4U^2C_y/\delta - r_y^2 &= 4U^2(V^2-1)/\delta,\\
4U^2D_\tau - r_\tau^2 &= U^2(4-k^2),\\
4U^2D_f - r_f^2 &= 4U^2(1-I^2),\\
4M^2D_m - r_d^2 &= 4M^2(1 + Hc^2 - 2acX - X^2).
\end{aligned}$$

The approximate root degrees are $25, 17, 13, 7$, and the corresponding formal square-root correction degrees are at most $3, 5, 5, 4$. To justify this calculation, scale every variable by an indeterminate u and work over the Laurent-series field of the rational function field in the original coordinates. Each radicand is an approximate-root square times 1 plus negative-degree terms, so the binomial series supplies its formal root with that leading sign. The delivered polynomial uses no series.

Eight conjugate factors have degree 25, four degree 17, two degree 13, and the last two degree 7. For the final two, the leading terms are U_0n and $U_0(n+c)$, where $U_0 = 16q_0^2a_0$. The product therefore has degree 308, and the clearing factor δ^8 adds 176. Thus $\deg N = 484$, and

$$(P_{540})_{540} = 2^{28}U_0^{16}Q_0^8V_0^8L_0^4F_{00}^2\eta_0^8n(n+c).$$

In particular the actual fixed-coefficient slice satisfies

$$[j^{240}w^{82}v^{94}h^{12}r^{34}c^{68}z^8n^2]P_{540} = 2^{530}3^{322}\beta^{240} > 0.$$

The selected exponents sum to 540, proving exact degree.

For the simpler companion only the auxiliary root is centered. Eight factors have degree 25 and eight have degree 17; the norm has degree $8 \cdot 25 + 8 \cdot 17 + 8 \cdot 22 = 512$. Now $U_0 = 8q_0^2a_0$, and

$$(P_{568})_{568} = 2^8U_0^{16}Q_0^8V_0^8L_0^8\eta_0^8.$$

Its selected coefficient is

$$[j^{256}w^{88}v^{104}h^{16}r^{32}c^{64}z^8]P_{568} = 2^{520}3^{344}\beta^{256} > 0.$$

The exact audits check these entire leading forms, all four square-defect identities, all 495 coefficients of the generic norm, all six fixed integer payloads against their finite reconstruction, and both literal DAGs. Finite modular comparisons are additionally retained as transcription diagnostics; the unbounded logical argument is the one above.

3 Removing unused radix positions in nine-variable equations

This section gives three complete polynomial constructions in nine strictly positive integers. The improvement has two parts: an exact translation of the finite coefficient layout, and a smaller evaluator for the translated polynomial. The translation is an existence-preserving change of the encoding; the evaluator change is an identity of polynomials. These two claims should be kept distinct.

Construction	Variables	Exact degree	Binary operations
Original31, simple majorant	9	1,472,168	420
Original31, complete radical majorant	9	997,810	1,057
Compact54, simple majorant	9	1,626,248	428

The respective incoming comparisons are the saturated H1868 simple form (1,474,688; 449 operations), its complete-radical form (999,518; 1,086), and the compact54 saturated form (1,628,768; 434). Thus the reductions relative to those immediate predecessors are respectively 2,520 degrees and 29 operations, 1,708 degrees and 29 operations, and 2,520 degrees and six operations. The 420-operation form also has lower degree and fewer operations than the 434-operation compact form, but it has a larger coefficient payload. These are improvements in stated measures within the inspected repository. They do not establish global minimality or reduce the number of variables below nine.

The first two predecessors are in `ConZFC_Saturated_Certificates_2026-10-24.zip`; the third is in `con-zfc-saturation-and-projection-2026-10-23.zip`. The repository snapshot is `7e533c33819ee1f703b69f2778a767cdc069940d`. The included source ledger identifies these exact archives by SHA-256.

3.1 A translation lemma for sparse coefficient layouts

Let $\rho_1, \dots, \rho_m$ be quadratic polynomials in ν variables, and suppose fixed coefficient polynomials $K_0(T), K_1(T), K_2(T)$, distinct nonnegative weights w_i, and distinct nonnegative target positions q_j satisfy

$$[T^{q_j}] \sum_{d=0}^{2} K_d(T)U_{\mathbf{p}}(T)^d = 2\rho_j(\mathbf{p}), \qquad U_{\mathbf{p}}(T) = \sum_{i=1}^{\nu} p_i T^{w_i}. \tag{1}$$

The equality is a polynomial identity in all source coordinates, including coefficients of monomials absent from the desired residual. It is stronger than a numerical identity at selected source tuples.

Lemma 3.1 (Translation of the layout)**.** *Let r, s be nonnegative integers such that $w_i \geq r$, $q_j \geq s$, and every occupied exponent e of K_d satisfies $e + dr - s \geq 0$. Set*

$$w_i' = w_i - r, \quad q_j' = q_j - s, \quad K_d'(T) = \sum_e [T^e] K_d(T)\, T^{e+dr-s}.$$

Then these are ordinary polynomial data and satisfy the same extraction identities with primes on w, q, K, U. If $W = \max_i w_i$ and $H = \max_d(\deg K_d + dW)$, their new values are $W' = W - r$ and $H' = H - s$.

Proof. As a temporary identity in Laurent polynomials, $U_{\mathbf{p}}' = T^{-r} U_{\mathbf{p}}$ and $K_d' = T^{dr-s} K_d$. Therefore

$$\sum_d K_d'(U_{\mathbf{p}}')^d = T^{-s} \sum_d K_d U_{\mathbf{p}}^d.$$

Coefficient extraction proves the claim. The hypotheses ensure that every exponent in the actual delivered data is nonnegative, so the construction contains no division by an unknown. Finally $\deg K_d' + dW' = \deg K_d + dW - s$, proving the height identity. □

Apply the lemma with $(r, s) = (1, 2)$. Both inherited layouts have minimum coordinate weight one, minimum occupied linear-kernel exponent one, and minimum occupied quadratic-kernel exponent zero. Their constant kernels begin above exponent two. Consequently the transformation reads

$$w_i' = w_i - 1, \qquad q_j' = q_j - 2, \qquad K_2' = K_2, \quad K_1' = K_1/T, \quad K_0' = K_0/T^2. \tag{2}$$

The last two expressions denote removal of initial zero coefficients; the JSON tables and the final circuits use nonnegative exponents only. No source coordinate, source equation, or numerical coefficient changes.

There is a useful restricted optimality statement. Among translations of this form with $r, s \geq 0$ and nonnegative resulting exponents, $r \leq 1$ because the minimum original weight is one, and $s \leq 2r$ because the quadratic kernel has a nonzero constant coefficient. Thus $(1, 2)$ is maximal in both parameters for these fixed layouts. This says nothing about changing the weights nonuniformly, changing the source residuals, or using a different encoding architecture.

The translated coordinate weights, in source-variable order, are

$$(0, 6, 60, 35, 15, 84, 19, 66, 42, 10, 33, 12, 95, 79, 28, 49,\\ 1, 108, 109, 111, 115, 90, 52, 54, 26, 100, 91, 8, 106, 36, 59).$$

For the original31 source, the target positions in row order are

$$(1139, 26, 1435, 1067, 1834, 1352, 1812, 612, 265, 1406, 1401,\\ 887, 335, 1585, 411, 671, 1137, 770, 1129, 1016, 1053, 83).$$

The source is the inherited positive system named `gapN86`, with 31 coordinates and 22 quadratic residuals. Its old coordinate maps are $P_{\text{old}} = P + \eta + \zeta$ and $N_{\text{old}} = N + 1$; its recorded elementary row operations are retained unchanged. The compact source is the separate `gapNq85` system. Its tables and whole compact54 coefficient tuple are kept separately; substituting individual coefficient ports between the two systems is not justified.

3.2 Why a coordinate may occupy position zero

The choice $w_i > 0$ was not needed by the positive-domain bit-mask argument. For distinct weights in $\{0, \ldots, W\}$, put

$$\ell = \sum_i B^{w_i}, \qquad N_0 = B^{W+1}, \qquad M_0 = N_0 - 1 - (b-1)\ell.$$

Assume b and B are powers of two, with $b \mid B$ and $b \le B$. The radix-B digits of M_0 are $B-b$ at selected positions and $B-1$ elsewhere. For $0 \le g < N_0$, binary disjointness $g \,\&\, M_0 = 0$ is therefore equivalent to

$$g = \sum_i z_i B^{w_i}, \qquad 0 \le z_i < b. \tag{3}$$

This digit argument includes position zero without any modification: the allowed bits there are precisely its bottom $\log_2 b$ bits. The recovered positive source coordinates are $p_i = z_i + 1$.

It is essential to check the next arithmetic interface rather than merely observe that the digit argument works. The pinned Lean source `Diophantine/Paper1982/NineRatio.lean`, retained in the packet, states `Jones1982.Nine.ratio_sound` and `ratio_complete` with the numerical hypotheses

$$R, N \ge 8, \quad 0 < b \le R, N,$$

and with the appropriate positive carrier, margin, central-binomial and power-of-two conditions. Neither theorem assumes $b \mid g$ or $b \mid R+1$. Hence the possible nonzero units digit introduced by (3) does not violate an inherited premise. This point is specific to the positive nine-variable interface; the signed Sun/Lucas construction has additional residue normalizations and is not changed here.

The all-one source tuple is not a common zero: its fifth residual is one, independently of the four coefficient ports. Thus every genuine source zero has at least one $p_i > 1$, and its packed g is positive even with a weight zero. For all sufficiently large binary b, its digits satisfy (3) and $g < bB^W$. The cofinal completeness property survives the translation.

3.3 The fully recalibrated original31 prefix

For this paragraph all layout data are the translated original31 data. Write K_d for their three kernels, and put

$$\kappa = 2134939, \qquad B = 2^\kappa b^2, \qquad T_* = \sum_{j=1}^{22} B^{q_j}, \qquad U_* = \ell + g.$$

Define

$$\begin{aligned}
G_* &= (K_2(B)U_* + K_1(B))U_* + K_0(B), \\
N_0 &= B^{116}, & M_0 &= N_0 - 1 - (b-1)\ell, \\
S_* &= g + N_0(2G_* + T_*), & T_c &= M_0 + N_0(B-2)T_*, \\
L &= 2bB^{1983}, & R &= L(S_* + T_c + 1) - S_* - 1, \\
N &= 2^{16012032}B^{1053}, & t &= bB^{115}.
\end{aligned} \tag{4}$$

Every exponent here is fixed. The kernel degrees are $(1406, 1726, 1636)$, so $W = 115$, $H = 1866$, and the largest target is $1834 = H - 32$. The leading kernel coefficients remain $-6, 2, 1$.

The inherited coefficient bound is $M = 2^{2134927}$, and $B = 4096Mb^2$. The exact artifact checker also verifies this bound on all actual fixed coefficients. For positive b, g, one has $U_* \ge B^{115}$ and $K_2(B) > B^{1636}/2$. Relative to $B^{1636}U_*^2$, the possible negative linear and constant contributions are bounded by $2M/B^{25}$ and $2M/B^{460}$. Thus $G_* > 0$ unconditionally. The low mask lies in $[0, N_0)$, so $S_*, T_c \ge 0$ and

$$R = (L-1)(S_* + T_c + 1) + T_c \ge L - 1 \ge 8.$$

Also $b \le R, N$ and $N \ge 8$.

On the soundness range $g < 4t$, let $Z_* = B^{1867}$. The same geometric-series estimate as in the inherited sparse-carry proof now gives

$$U_* < 6bB^{115}, \qquad \frac{2G_*}{Z_*} < \frac{86Mb^2}{B-1} < \frac{1}{32}.$$

The two target-mask bounds are unchanged because their headroom $H - Q = 32$ is unchanged. Together with $g < N_0$ and $M_0 < N_0$, they give

$$S_* + T_c < N_0 Z_* = B^{1983} < L.$$

Each source convolution coefficient has magnitude at most $993Mb^2 < B/4$. The retained signed-digit carry lemma therefore applies with exactly the same strict inequalities. It identifies $S_* \,\&\, T_c = 0$ with a positive common source zero.

For completeness, the threshold must be recalculated. Write $b = 2^k$, $B = 2^v$, where $v = \kappa + 2k$, and $L = 2^u$. There are 116 low radix digits, 31 selected low positions, and 22 high mask digits. Consequently

$$\operatorname{pop}(T_c) = 138v - 31k - 22, \qquad u = 1 + k + 1983v.$$

The two-block population identity gives

$$\operatorname{pop}(R) = u + \operatorname{pop}(T_c) - \tau, \quad \tau = \operatorname{pop}(S_*) + \operatorname{pop}(T_c) - \operatorname{pop}(S_* + T_c) \geq 0,$$

with $\tau = 0$ exactly for binary disjointness. Now

$$\begin{aligned} u + \operatorname{pop}(T_c) &= 2121\kappa + 4212k - 21 \\ &= 2\big(16012032 + 1053(\kappa + 2k)\big) = 2\log_2 N, \end{aligned}$$

since $15\kappa - 21 = 32024064$. Legendre's formula therefore yields

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_* \,\&\, T_c = 0. \tag{5}$$

The argument includes $b = 1$. At that value $M_0 = N_0 - 1$ and $0 < g < N_0$, so disjointness fails. There is no exceptional accepted assignment at the endpoint.

For the compact54 variant replace the source tables and the whole native coefficient tuple, and use

$$\kappa = 864026, \quad H = 2080, \quad Q = 2048, \quad L = bB^{2197}, \quad N = 2^{6480184}B^{1160}, \quad t = bB^{115}.$$

The kernel degrees are $(1625, 1940, 1850)$, the dominance gaps are 25 and 455, and the coefficient bound is 2^{864014}. Its population calculation is $u + \operatorname{pop}(T_c) = 2335\kappa + 4640k - 22 = 2\log_2 N$. Thus the same proof applies. The exact source audit checks all 22 identities separately for each of these two genuinely different sources.

3.4 The actual nine-variable polynomials

The unknowns are

$$b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}.$$

All other names in this subsection are abbreviations, not quantified variables. Use either certified prefix above and define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2 U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R+1, & W_1 &= bw,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1),\\
m &= K^2(4t-g) - 16t(C-KY)^2, & \Omega &= Jm,\\
E_0 &= J(n + W_1^2) + Q, & V_0 &= PK,\\
a_0 &= (P^2-1)K^2 + 1, & D &= (A^2-1)C^2 + 1,\\
r_0 &= iJC^2, & F &= (A^2-1)Dr_0^2 + 1,\\
\Lambda &= CD + 1 - A, & G &= A + F\Lambda,\\
Z &= B_1 + (j-1)C, & & \\
H_1 &= C + FZ, & I &= (G^2-1)H_1^2 + 1.
\end{aligned}$$

For either simple construction set

$$a_1 = DFI, \qquad V_1 = ADr_0 G H_1.$$

For the complete-radical construction set

$$\sigma = 1024C^5 i, \qquad a_1 = \sigma^2 DFI, \qquad V_1 = \Pi(A, C, i, Z),$$

where Π is the fully specified inherited 333-term integer polynomial in `data/saturated_corrected_P.json`. Its exact identity certificate is retained; it is not an unspecified choice of a majorant. Finally set

$$\Theta = E_0 - \Omega(V_0 + V_1), \qquad \boxed{\left(\Theta^2 - \Omega^2(a_0 + a_1)\right)^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{6}$$

The three literal division-free DAGs, all fixed numerical leaves, and finite coefficient reconstruction recipes are included in the packet.

Theorem 3.2 (Preservation of the consistency encoding)**.** *The three positive-domain equations* (6) *have a zero if and only if their respective inherited positive source systems have a common zero. Composing with the retained native-source and machine/refutation calibration gives the same ordinary theorem formalizable in ZFC:*

$$\mathrm{ZFC} \vdash \left(\nexists (b, g, h, s, w, C, i, j, n) \in \mathbb{Z}_{>0}^9 : \mathcal{F} = 0\right) \iff \mathrm{Con}(\mathrm{ZFC}).$$

Proof. The preceding arguments establish all the numerical prefix hypotheses, its soundness range, cofinal completeness, and the exact central-binomial calibration. The source zero set is unchanged by Lemma 3.1. The saturated positive ProductPell finalizer is retained unchanged.

For clarity, its proof order matters. Unconditional strict bounds $V_0 > \sqrt{a_0}$ and $V_1 > \sqrt{a_1}$ first force $m > 0$ from a norm zero. They then give the carrier range and strict separation of the two radicands. A nonzero rational sum of their signed square roots forces each root to be rational, hence integral. This supplies the square conditions and $J \mid Q$. The saturated ProductPell theorem recovers $C = \psi_A(2R+1)$, after which the original numerical ratio theorem recovers the binary base and (5). No inference about square roots is used prematurely to obtain the margin. Conversely a source zero supplies fresh ratio and Pell witnesses, and the inherited

integral choice of $n > 0$ makes one norm factor vanish. The recorded soundness and saturation proofs prove these steps in full.

For the simple majorant the identity

$$(ADr_0)^2 - DF = D(Dr_0^2 - 1) > 0$$

implies the required strict bound after multiplication by $G^2H_1^2$. For the complete majorant, the exact polynomial $\Pi^2 - \sigma^2 DFI$ has 36,309 strictly positive coefficients after $A = x + 3, C = y + 1, i = u + 1, Z = v + 1$, including a positive constant. The translated polynomial Π itself also has positive coefficients. Thus the same majorant premise holds uniformly. These are retained algebraic certificates, rechecked exactly on the present parameters.

All new steps are elementary finite polynomial identities, integer digit arguments, or arithmetic statements with the same formalizable inherited interfaces. The packet does not supply a new end-to-end Lean/Rocq proof or an exported object-language ZFC derivation; its calibration dependencies are explicit rather than inferred from finite testing. □

3.5 Exact degrees and exact arithmetic-circuit checks

For the original31 prefix,

$$r = \deg R = 7931, \quad \nu = \deg N = 2106, \quad a = \deg A = r + 4\nu + 2 = 16357.$$

For compact54 these values are $8787, 2320, 18069$. The positive leading source term is $R \sim 2^{\lambda+1} b B^{2E-1}$, where $(E, \lambda) = (1983, 1)$ or $(2197, 0)$. Thus these bounds are attained on the actual fixed-coefficient instances. Write $\delta_0 = V_0^2 - a_0$, $\delta_1 = V_1^2 - a_1$, and

$$B_s = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1).$$

The exact cancellation-resolved identity is

$$\mathcal{F} = B_s^2 + 4\Omega^2 V_0 V_1 B_s + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1). \tag{7}$$

One has

$$\deg\Omega = 5a + 2W + 3, \quad \deg V_0 = 3a + r + 1, \quad \deg\delta_0 = 3a + r + 2, \quad \deg E_0 = a + 4.$$

For the simple majorant, $\deg V_1 = 18a + r + 24$ and $\deg\delta_1 = 34a + 2r + 48$. The uniquely dominant term of (7) is $\Omega^4\delta_1^2$, giving

$$\deg\mathcal{F} = 88a + 4r + 8W + 108.$$

For the complete majorant, exact expansion gives

$$\deg V_1 = 18a + r + 30, \qquad \deg\delta_1 = 18a + 2r + 60.$$

Their unique leading monomials before composition are

$$1048576A^{18}C^{24}i^6Z, \qquad 624687054848A^{18}C^{48}i^{12}Z^2.$$

The four contribution degrees in (7) are $948768, 997810, 989850, 752426$. The second is uniquely maximal, and its leading coefficient is positive, giving degree 997810. Merely propagating maximum degrees through the unresolved norm would not establish these exact values.

The evaluator uses one addition chain for every required power of B. The selected circuits evaluate ℓ and T_* as sums of shared powers, and evaluate the three coefficient kernels in Horner form. This saves 26 multiplications for the original31 forms compared with using Horner form for both masks; it saves 25 for the compact profile. The common power chain is a finite table

of equalities $e = u + v$; its use costs one multiplication per live chain step. Fixed numeral construction is excluded from this convention, but all live binary additions, subtractions, squares, and multiplications by numerals are counted.

The supplied verifier does not import the builder. It proves 54 or 55 exact polynomial identities at the named registers of each DAG, including all complete-majorant coefficients. It separately checks the variable set, acyclicity, liveness, operation counts, fixed-leaf recipes, and the final output. A second independently implemented sparse ring checks all 22 source identities including absent monomials, all 20,466 original31 and 22,450 compact convolution interactions, the coefficient bounds, the positive majorant certificate, the population exponents, and the resolved degree. Eight complete native coefficient byte strings are reconstructed from the retained literal finite inputs. These checks support exact algebra and the numerical instances; they do not experimentally test consistency or substitute for the unbounded semantic calibration.

The coefficient-size tradeoff is material. The original31 tuple occupies 533,099 raw bytes in its four ports, versus 215,492 for compact54. Its twelve live fixed leaves occupy 4,001,391 raw bytes and 53,739 gzip bytes, versus eleven leaves occupying 1,565,018 raw bytes and 23,481 gzip bytes for compact54. The larger live-leaf total includes the fixed powers of two and derived coefficient products. These are distinct measures from circuit length and from the size of the entire research archive.

3.6 Further questions suggested by the construction

The uniform translation is maximal for these fixed supports, but a joint optimization of coordinate weights, target positions, row operations and evaluator structure may improve all three circuits further. Such a search should allow weight zero from its beginning and score the final polynomial rather than only the convolution height. Another concrete target is a Lean proof of the general layout-translation and zero-position mask lemmas, composed with the existing ratio interfaces. Finally, the exact register certificates suggest a small checked circuit verifier that accepts the whole finite DAG and the fixed numerical leaves without trusting the arithmetic-circuit producer. None of these questions entails a claim that nine variables are necessary or that a different direct proof encoding cannot give a simpler equation.

4 A direct quartic with 495 nonnegative unknowns

The direct branch encodes finite derivations in the repository's calibrated numerical calculus C_{222}. Its immediate parent is the incoming synchronized-certificate construction with 522 nonnegative unknowns. Combining its arithmetic certificates with branchwise projection of rule labels gives the following fully materialized equation.

Measure	Synchronized parent	Projected equation
Nonnegative unknowns	522	495
Quadratic residuals	400	391
Exact total degree	4	4
Expanded monomials	20,225	15,377
Maximum absolute coefficient	83,900,628	83,867,632

The actual polynomial is

$$P_{495}(z_0, \ldots, z_{494}) = \sum_{j=1}^{391} R_j(z_0, \ldots, z_{494})^2.$$

Every coefficient of every quadratic R_j, and every collected coefficient of P_{495}, is supplied as an ordinary integer in the literal artifacts. The notation does not quantify additional registers.

The source equations below explain the construction; all their powers, binomials, factorials, and bounded quantifiers have been removed from the delivered quartic.

This is a measured improvement of the direct, modest-coefficient quartic branch. It is not a claim that 495 is an optimal variable count. The construction does not encode a run of K215 and does not use Jones's universal polynomial. Its ordinary consistency theorem imports the identified C_{222} calibration; it is not a new end-to-end proof-assistant certificate.

4.1 Project the premise and conclusion labels first

Write

$$I(a,b) = 2(a+b)(a+b+1) + 4b, \qquad A(a,b) = I(a,b) + 2.$$

The calibrated source has twelve branches. Its truth and axiom branches have premise labels $p = q = 0$ and conclusion label $v = h_b(a,b,c,d)$, with truth represented by $h_b = 0$. The exact nine nonzero conclusion polynomials are the inherited numerical constructors. The two inference branches admit the parameterizations

$$(p,q,v) = (a, I(a,b), b), \qquad (p,q,v) = (b, 0, A(a,b)).$$

For modus ponens these parameters are the old first premise and conclusion; for generalization they are its generalization parameter and premise. Every old instance therefore has a new instance, and substitution recovers an old instance from every new one. Since existential quantification distributes over a finite disjunction, unused parameters in an unselected branch impose no compatibility condition.

There is one further local saving. The bundle branch's conclusion is independent of its fourth parameter. Its guard

$$g^2 = (d_{\mathrm{old}} + 1)^2, \qquad g = (c-a)(c-b), \qquad d_{\mathrm{old}} \geq 0,$$

can be replaced by $g^2 = d + 1$, $d \geq 0$. Both existential conditions mean $g \neq 0$. An old witness gives $d = d_{\mathrm{old}}^2 + 2d_{\mathrm{old}}$; a new witness gives $d_{\mathrm{old}} = |g| - 1$. The fourth parameter in another selected branch retains its original meaning.

Fix outer parameters $t, m \geq 1$ and $x, y, M_0 \geq 0$. Set $M = M_0 + 1$ and $w(j) = 1 + Mj$. For each branch let

$$e_b = S'w(v_b) - Sw(p_b)w(q_b).$$

Let $e_0, \ldots, e_8$ be the truth and eight unguarded axiom residuals; let $e_{\mathrm{MP}}, e_{\mathrm{GEN}}, e_{\mathrm{B}}$ be the two inference and bundle residuals. Put $G = ((c-a)(c-b))^2 - d - 1$. The complete row equation is

$$\begin{aligned} 0 = {} & (e_0 \cdots e_8 e_{\mathrm{MP}} e_{\mathrm{GEN}})^2 \, (e_{\mathrm{B}}^2 + G^2) \\ & + \big[x - S - (1 + (i+1)y)u\big]^2 + \big[x - S' - (1 + (i+2)y)v\big]^2. \end{aligned}$$

Every summand is nonnegative. Consequently this equation is equivalent to the two beta reconstructions together with one valid selected rule's accumulator identity. Only eight bounded row coordinates remain:

$$(a,b,c,d,S,S',u,v), \qquad (a,b,c,d,S,S',u,v) \leq (m,m,m,m,y,y,x,x)$$

coordinatewise. The implementation also shares $S'M$ in $S'w(h) = S' + (S'M)h$ and squares the product of the eleven unguarded residuals once. These are exact polynomial identities, not probabilistic fingerprint equalities.

4.2 A small circuit bounds all labels

At $a = b = c = d = m$, a literal certificate bounds every constructor by a fixed power of $R_0 = m + 2$. Assign exponent one to m, exponent $\lceil \log_2 c \rceil$ to a positive numeral c, exponent zero to zero, and propagate

$$e_{U+V} = \max(e_U, e_V) + 1, \qquad e_{UV} = e_U + e_V.$$

Since $R_0 \geq 2$, induction proves $U \leq R_0^{e_U}$ at every gate. All constructor coefficients are nonnegative, so the diagonal evaluation also bounds the labels when the four parameters are merely at most m. The largest certified exponent is $3,445,903$; use

$$h(m) = (m+2)^{4,194,304}, \qquad M = (h(m)+1)^t.$$

The fixed power defining h takes 22 squarings. This high semantic degree does not raise the quartic degree of its arithmetic graph.

Here is the exact carrier argument. For the t selected rule instances define

$$B_0(X) = \prod_{i<t}(1 + v_i X), \qquad B_1(X) = (1+X)\prod_{i<t}(1 + p_i X)(1 + q_i X).$$

The endpoints $S_0 = M + 1$, $S_t = 1$ and the accumulator identities imply $B_0(M) = B_1(M)$. All states are positive, by induction from the first endpoint and positivity of the label weights. With $h = h(m)$,

$$B_0(M) \leq (1 + hM)^t < ((h+1)M)^t = M^{t+1}.$$

Thus $\deg B_1 \leq t$, since a nonnegative integer polynomial of greater degree would have value at least M^{t+1}. Both polynomials now have coefficient sum at most $(h+1)^t = M$, and constant coefficient one. Each coefficient is therefore strictly smaller than M. Uniqueness of base-M digits proves $B_0 = B_1$. Comparing the multiplicities of the distinct factors $1 + jX$, $j > 0$, gives the exact Horn balance for target label one. The inherited closure-separation argument turns that balance into a finite derivation, without excluding cycles or repeated premises.

Conversely, expand a finite derivation to a finite tree. Choose t for its rule occurrences and choose m afterward to bound their four parameters, including each new bundle slack $g^2 - 1$. Products of the weights of outstanding obligations give the required integral accumulator trace. The independent parameter m avoids any need to pad the derivation until its horizon bounds its own labels.

4.3 Compile and contract

Choose y above the finitely many states and divisible by $t!$, then choose a beta code x for the trace. Its natural quotients are at most x. Conversely, the bounds $S, S' \leq y$ make both reconstructions canonical remainders, since $y < 1 + (i+1)y$.

A second literal majorant certificate uses base $2 + t + x + y + m + M_0$ and fixed exponent $8,388,608$. It bounds both nonnegative sides of the row equation at all upper row bounds, as well as the index and the individual bounds. The inherited fixed-vector CRT theorem then applies to the eight columns. Its factorial factors cancel because the CRT moduli are one modulo the chosen common factorial.

The resulting finite arithmetic circuit makes 11 binomial requests and 24 positive-power requests, with 19 distinct Pell indices and 10 distinct power bases. The shared binomial radix and common Pell-frame theorems apply to this finite family unchanged. In soundness one identifies the power graphs first, the binomial digits second, and the factorial/CRT semantics afterward. In completeness one chooses the intended acyclic semantic values first and the shared auxiliary certificates afterward. This order avoids circular justification of the shared radix or frame.

The emitted parent of the final quartic has 823 natural variables and 719 quadratic residuals. Exact nonnegative monic substitutions eliminate 328 further variables and residuals. The inverse substitutions have nonnegative integer coefficients, and the retained residuals remain quadratic. Thus both existence implications hold over the naturals. The independent replay verifies all 328 substitutions and the complete 15,377-term sum-of-squares expansion.

5 Recover nonnegativity from the equations themselves

Converting every natural coordinate independently would waste many variables. The displayed equations already force most coordinates to have the required signs. A finite certificate shows that guarding only 170 of the 495 coordinates suffices. The main additional argument uses the common Pell frame before invoking any exponentiation semantics.

5.1 Local sign implications

Suppose a literal residual can be written

$$A(\mathbf{z})z_i = B(\mathbf{z}),$$

where neither A nor B involves z_i. If previously established facts prove $A > 0$ and $B \geq 0$, then $z_i \geq 0$; if $B > 0$, then $z_i > 0$. The certificate uses this elementary implication and the weakening $z_i > 0 \Rightarrow z_i \geq 0$. Its sign checks allow even powers without assuming a sign for their bases. Positive integer monomials can also pay an explicitly bounded negative constant term.

Starting with the 170 declared guards, 531 such checked implications prove 447 of the 495 coordinates nonnegative. Crucially, this first stage proves all prerequisites of the following common-frame argument. No T or Pell abscissa whose sign is recovered by that argument is used prematurely.

5.2 One frame proves 29 further signs

The exact raw compiler identities restore polynomially from the final 495 coordinates. Let W, A, Z be its common frame, and let (n, k, m) range over the finite power requests. The frame identities are

$$W = 2 + \sum_{\xi \in \mathcal{A}} \xi, \qquad A^2 = 1 + W^3(W+2)Z^2.$$

Here $\mathcal{A}$ is the finite set of distinct base, exponent, and output atoms in the circuit; distinct atoms with equal values are still counted separately. The exact list is supplied. Before the new argument, all these atoms are nonnegative, every base n is strictly positive, $A \geq 0$, and $Z \geq 1$. These facts are checked through the literal restorations, including the factorial output used as a power base. Consequently

$$W \geq 2, \qquad 1 \leq n \leq W, \qquad A > W^2.$$

The last strict inequality follows immediately by squaring; no Pell classification or already-established power semantics is needed.

Each of the ten distinct base moduli satisfies

$$T_n = 2An - n^2 - 1.$$

Therefore $T_n \geq 2A - W^2 - 1 > 0$. For each power request the exact oriented congruence is

$$X + Yn = YA + m + T_n q,$$

and the first sign stage has already established $Y, m, q \geq 0$. Since $A - n > 0$, rearranging gives

$$X = Y(A - n) + m + T_n q \geq 0.$$

This recovers the signs of all 19 distinct index abscissae. After adding the ten modulus signs and these 19 abscissa signs, 21 further local implications prove all 495 coordinates nonnegative.

The order matters: base positivity precedes the frame argument, modulus positivity precedes abscissa recovery, and full natural-domain arithmetic semantics is invoked only after every coordinate has been recovered. This separates a short ordered-ring proof from the much stronger Pell-index argument used later to identify actual powers.

5.3 A quartic in 977 unrestricted integers

For an integer x, the classical three-square theorem gives

$$x \geq 0 \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Z} : \quad x = u^2 + v^2 + w(w + 1).$$

Soundness is immediate. For completeness, $4x + 1$ has a representation by three squares; modulo four exactly one square is odd. Writing that odd integer as $2w + 1$ and the even integers as $2u, 2v$ proves the formula.

Adjoin this quadratic identity for each of the 170 guarded coordinates. The other coordinates require no separate representation, because the checked domain-recovery theorem forces their nonnegativity. Keeping the original coordinates at this stage preserves degree two in every residual. The intermediate unrestricted-integer system therefore has

$$495 + 3 \cdot 170 = 1005 \quad \text{variables}, \qquad 391 + 170 = 561 \quad \text{quadratic residuals}.$$

Twenty-eight further signed monic substitutions preserve quadratic degree and existence over the integers. The final literal equation has

977 unrestricted integer unknowns,	533 quadratic residuals,	$\deg P_{977} = 4$,
24,632 expanded monomials,	$\max \lvert\text{coefficient}\rvert = 83,869,250$.	

The comparison with the synchronized parent's 2088-integer quartic is a reduction of 1111 integer variables, or approximately 53.2%, while retaining degree four. This is a comparison within the direct branch; the repository's much smaller-variable high-degree constructions remain different tradeoffs.

Every integer zero of P_{977} restores the guarded 495-coordinate system, hence a completely nonnegative zero of P_{495} and a calibrated finite refutation. Conversely, a refutation gives a natural zero, three-square witnesses for the 170 guards, and then a zero of the contracted integer system. Composing with the named ordinary source calibration and negating yields the desired ZFC-provable biconditional between absence of integer zeros and Con(ZFC).

5.4 What the finite verifier establishes

The domain checker is independent of the sign-discovery code. It rederives every local implication from a literal quadratic residual and checks 1535 raw polynomial restorations. Of the raw equations, 1040 restore to zero identities and 391 restore to final residuals. It also binds all 1243 raw arithmetic-gate metadata records to their exact defining residuals before expanding them. Finally it checks the frame norm and sum, ten modulus identities, and all 24 oriented power identities. The ordinary inequalities above then justify its 29 additional sign conclusions.

Corruption controls require rejection after a gate operand is changed, a guard is deleted, a pivot orientation is reversed, a frame conclusion is omitted, or a modulus identity is replaced. Normal and optimized Python runs perform the checks explicitly. These finite verifications certify the formulas, transformations, and coefficient data; they do not replace the unbounded source proof or decide whether a refutation exists.

6 Logical dependencies and reproducibility

6.1 The direct branch reaches consistency without a machine run

The direct source is the numerical calculus C_{222}, with distinguished contradiction code one. Two inherited theorems identify its consistency meaning [2]. In the retained source, the first theorem is

```
KEndpoint.c222_iff_zf_refutation
```

and states that derivability of the distinguished code in C_{222} is equivalent to a finite ZF refutation. It appears before the machine-parameter section of `KEndpoint.lean`; it has no machine, tape, simulation, boot, coverage, or halting premise. Its upper direction uses the interpretation of the numerical rules, and its lower direction uses the established semantic completeness chain of the calculus.

The second theorem is

```
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

Its forward implication uses theory inclusion. Its reverse implication uses the constructible model of a model of ZF, together with first-order soundness and completeness. These are the inherited ordinary calibration results needed for the direct polynomial. Their complete immediate declaration files and their exact source hashes are retained under `direct/provenance/upstream/calibration/`.

The retained calibration revision is `08200eb10fde2db408c69dffd840d6920a569b1f`. This is intentionally distinct from the comparison snapshot in the title: the incoming parent pinned those earlier theorem bodies, and this packet preserves its pins instead of silently replacing them with a later branch. The ordinary CRT, binomial, factorial, power, and shared-frame proofs are also retained.

Thus the direct branch encodes derivations in a numerical calculus and does not simulate K215 or substitute parameters into Jones's universal polynomial. Some files in the inherited Lean import graph also support machine constructions. No claim is made that the entire import graph is free of machine-related modules.

6.2 The compact branches preserve their native calibrations

The fourteen-integer branch retains the coherent source-34/source-71 coefficient tuple and its signed native kernel, ordinary compiler calibration, and K215/refutation endpoint. The new proof establishes a two-way existence equivalence to that kernel, including the restoration of its strong variables. The six fixed coefficient values are finite literal integers in the packet. Their finite reconstruction is checked; their semantic interpretation comes from the inherited compiler proof.

The nine-variable branch likewise keeps its two coherent native tuples separate: the original31 `gapN86` source and the compact54 `gapNq85` source. It imports the positive sparse-carry encoding, the numerical ratio theorem, the saturated ProductPell theorem, and the native source/refutation calibration. The new proof checks the exact premises affected by the origin shift and recalibrates the threshold; it does not infer those unbounded implications from a few sampled values.

The conversion from a nonnegative integer to three integer witnesses is classical. For $x \in \mathbb{Z}$, the exact characterization $(\exists u, v, w \in \mathbb{Z} :\ x = u^2 + v^2 + w^2 + w) \iff x \geq 0$ is also recorded in the Archive of Formal Proofs formalization of the three-square theorem [3]. Here it is used only after the 170-coordinate domain recovery theorem has been established.

6.3 Four distinct levels of evidence

It is useful to separate what has actually been delivered.

1. **Explicit equations.** All seven final polynomials are literally determined. The large-coefficient forms have acyclic circuits and materialized fixed leaves. Both quartics have complete expanded coefficient lists as well as quadratic sum-of-squares presentations.
2. **Ordinary mathematical proofs.** The article proves the new source changes, guard and radical arguments, domain recovery, and witness restorations. The retained sources state the precise inherited semantic calibrations used in composition.
3. **Exact finite certificate checks.** Independent arithmetic implementations check relevant identities, source layouts, leading terms, sign implications, contractions, and collected coefficients. Rebuilding recovers the delivered equation files byte for byte.
4. **Formal proof artifacts.** Existing pinned Lean declarations are retained as dependencies. This work has not built a new kernel-checked theorem for the seven final polynomials or exported a derivation in the sealed object-language ZFC system. Those remain concrete formalization tasks, not claims supported by the Python checks.

The internal mathematical reviews in the packet were conducted by separate AI work streams. They are useful adversarial checks of the construction, not external peer review. The finite verifiers themselves are short ordinary programs and remain subject to inspection; their rejection controls exercise actual failure paths, including when Python assertions are disabled.

6.4 Reproduce all finite results

Python's standard library is sufficient. From the extracted packet root, run

```
python3 verify_all.py --output verification/rechecked.json
python3 -O verify_all.py --output verification/rechecked_O.json
```

The driver copies the three components to a temporary directory, checks the delivered artifacts before rebuilding them, and compares all reconstructed final equations with the original bytes. It leaves the delivered equation and certificate files unchanged. The two recorded root receipts are `verification/ROOT_NORMAL.json` and `verification/ROOT_OPTIMIZED.json`.

The finite checks include the following substantive obligations:

- All 495 coefficients of the four-root norm, four exact root-defect identities, both degree-leading forms, all six fixed coefficient values, and the literal fourteen-variable circuits.
- All 22 source extraction identities for each positive packing, including absent monomials; 20,466 and 22,450 convolution interactions; all eight native coefficient ports; and 54 or 55 exact register identities per final nine-variable circuit.
- The complete 36,309-term positivity certificate for the full radical majorant and the cancellation-resolved degree calculation.
- The direct source constructor identities, both majorants, arithmetic certificate interfaces, 328 nonnegative contractions, and the full quartic expansion.
- The 170-guard sign certificate, 1,535 raw coordinate restorations, 1,243 gate bindings, the common frame and power identities, 28 signed integer contractions, and all 24,632 final quartic coefficients.
- Rejection after a guard, gate operand, pivot orientation, frame conclusion, modulus identity, restoration coefficient, or expanded coefficient is corrupted.

Finite modular comparisons retained by the fourteen-variable audit are transcription diagnostics. They are not substituted for exact symbolic identities, degree arguments, or unbounded semantic proofs.

7 Repository integration and next mathematical targets

7.1 Files to treat as authoritative

The root `README.md` maps each construction to its literal file. For P_{540} and P_{568} use the JSON circuits immediately under `integer14/`; for the three nine-variable forms use `nine/equations/`. The natural quartic is `direct/equations_v3/cap64/equation.json`, and the integer quartic is `direct/integer977/equation.json`. Each quartic directory also contains `expanded_polynomial.json`. The accompanying schema note specifies coefficient signs, variable indices, fixed leaves, and monomial conventions.

The archive keeps original source notices and the repository's MIT No Attribution license. The comparison ledger records the inspected main commit, incoming archive names, Git object identifiers, and downloaded SHA-256 values. Original archive filenames containing a later date have not been renamed. The packet-wide SHA-256 manifest binds the final files; it is an integrity record, not a mathematical proof.

No existing numbered repository construction is silently replaced. A new entry should state the domain and all selected size measures explicitly. The word "simpler" should be qualified by those measures. In particular, the 420-operation original31 form uses a larger native coefficient tuple than the 428-operation compact54 form. Keeping both makes that choice visible instead of hiding the coefficient cost in fixed leaves.

7.2 Research questions made concrete by this packet

Can the four-root mechanism keep thirteen variables? The gain at fourteen variables comes from retaining the guard root. Reabsorbing it without reintroducing the removed amplifier is a precise target. Any proposed fifth-root fusion must handle dependent square classes and both branches $cT - R = 0$ and $cT - R \neq 0$; a generic norm identity alone will not establish rational extraction.

Can a better center reduce degree below 540? The exact defect identities explain which factors remain expensive. They provide a starting point for a systematic search over polynomial root approximations. A candidate center must retain the CRT congruences, integrality of the final witness, and control of exceptional branches. Degree estimates must be resolved after cancellation on the authentic fixed-coefficient slice.

Can source layout and evaluator be optimized jointly? The uniform radix shift is maximal in its restricted class. That leaves nonuniform weight changes, target changes, source row transformations, and globally shared addition chains. A useful search objective is the actual final circuit and degree, with coefficient bytes reported alongside them. Optimizing convolution height alone can miss an evaluator saving.

How few signs really need guards? The set of 170 guards is sufficient, not proved minimal. A smaller seed set for the present sign rules, additional multi-row inequalities, or stronger frame consequences may save more variables. The explicit certificate separates discovery from verification, so a heuristic search may propose guards while an independent checker establishes the result. The final count should be measured after signed contractions, since a guard's cost can depend on the resulting elimination opportunities.

Can the direct source be reduced before arithmetic compilation? The successful projection removes deterministic labels before applying CRT. Further source changes should preserve exact parameter dependencies, the finite-tree completeness argument, and the carrier

coefficient bound. Sharing expressions across mutually exclusive rule branches may give additional savings without changing the underlying numerical calculus.

Can the whole endpoint be checked in one proof kernel? A particularly valuable next deliverable would connect the emitted literal quartic, its coefficient expansion, the domain certificate, and the inherited C_{222} consistency endpoint in a single kernel-checked theorem. The certificate formats and pinned declarations in this packet make that task specific. It requires formalizing the universal arguments, not merely embedding successful Python exit codes in a theorem statement.

References

[1] Vladimir Reshetnikov and ProveIt contributors. *ProveIt: Hilbert's tenth problem, Diophantine consistency forms, and incoming proposals.* Comparison commit `7e533c33819ee1f703b69f2778a767cdc069940d`. https://github.com/VladimirReshetnikov/ProveIt/tree/7e533c33819ee1f703b69f2778a767cdc069940d/docs/incoming. The accompanying comparison ledger records individual archive hashes.

[2] ProveIt contributors. *KEndpoint and ConstructibleChoice calibration sources.* Retained revision `08200eb10fde2db408c69dffd840d6920a569b1f`. https://github.com/VladimirReshetnikov/ProveIt/blob/08200eb10fde2db408c69dffd840d6920a569b1f/Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint.lean. Exact immediate copies and SHA-256 pins are included in the packet.

[3] Anna Danilkin and Loïc Chevalier. *Three Squares Theorem.* Archive of Formal Proofs, 3 May 2023. https://isa-afp.org/entries/Three_Squares.html.

[4] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher. *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization.* arXiv:2506.20909v3, 22 October 2026. https://arxiv.org/html/2506.20909v3.